

How Robust Are LLMs to Vietnamese Dialects?

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Abstract

Large Language Models (LLMs) are typically evaluated on standard written Vietnamese, yet everyday communication frequently involves regional dialects that preserve meaning but differ in surface form. Existing Vietnamese dialect work largely addresses this issue through dialect-to-standard normalization instead of measuring how the model fails under Vietnamese dialectal inputs. To address this gap, we present the first study that systematically evaluates the robustness of LLMs to Vietnamese dialect variation across multiple tasks, quantifying dialect-induced performance degradation and analyzing failure patterns behind it. We introduce **Vialect-Bench** (Vietnamese Dialects Benchmarking), a benchmark containing 2.4K+ human-annotated minimal pairs in standard Vietnamese and five regional dialects. Our benchmark covers MCQA, NLI, NER, and sentiment classification, enabling controlled analysis of dialect-induced performance drops and error patterns across dialects, tasks, and models. Beyond evaluation, we propose a test-time adaptation method that improves robustness under dialectal variation while preserving dialect-specific meanings without fully normalizing inputs into standard Vietnamese.

1 Introduction

[Minh Tran: High impacted paper has a strong abstract, introduction, figure, and conclusion as much as experimental design. For our paper, the introduction should tell this story:

1/ LLMs are usually eval on standard Vietnamese: BertPho, HutiuBert,...

2/ Real Vietnamese users write in dialectal forms

3/ Existing work mainly normalize the dialects but does not tell us whether LLMs themselves are robust

4/ Therefore we build a controlled benchmark to measure semantic consistency across dialects

5/ We show where models fail and how robustness can be improved.] Large Language Models

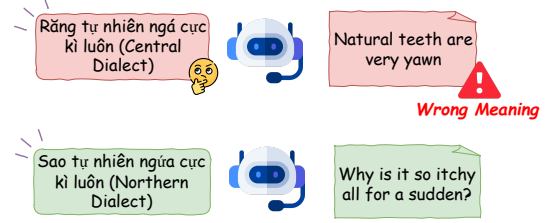


Figure 1: **Overview of meaning-preserving in NLP models under Vietnamese Dialects.** We evaluate whether LLMs produce consistent outputs when semantic meaning is preserved but dialectal forms change [Minh Tran: Figures should stand their own, ex figure 1 can explain the problems, figure 2 explain method, 3,4 explain analysis]

(LLMs), typically built on Transformer architectures (Vaswani et al., 2017), have achieved state-of-the-art performance across a wide range of natural language processing tasks (Zhao et al., 2026). However, most LLM evaluation protocols implicitly assume that users interact with models through standardized written language. This assumption is problematic for Vietnamese, where everyday communication frequently involves substantial regional variation. Vietnamese is commonly associated with three macro-dialectal groups—Northern, Central, and Southern—but dialectal diversity also appears at a finer province-level scale. Since standard Vietnamese is largely based on the Northern variety, many Vietnamese NLP models and benchmarks are trained and evaluated predominantly on standard written Vietnamese. As a result, model performance may degrade when inputs contain regional lexical, phonological, or syntactic variants, leading to misunderstanding, or hallucinated outputs.

Research Gaps and Novelty. Existing works (Le and Luu, 2023; Anh Ta et al., 2026) have provided important evidence that dialectal Vietnamese creates real challenges for NLP systems. Le and Luu (2023) introduce a parallel corpus for

Vietnamese Central–Northern dialect text transfer. Their work demonstrates that Central dialectal expressions can degrade downstream systems such as machine translation and text-image retrieval, while fine-tuned dialect transfer modules can improve performance without retraining the downstream models. Similarly, ViDia2Std (Anh Ta et al., 2026) introduce ViDia2Std, a nationwide Vietnamese dialect-to-standard corpus covering all 63 provinces, and show that normalization improves downstream tasks such as translation and sentiment analysis. While these studies provide valuable foundations for Vietnamese dialect-aware NLP, they primarily focus on building a corpus for dialect-to-standard normalization. However, it remains unclear whether modern LLMs are inherently robust to meaning-preserving dialectal variation. Existing datasets and methods evaluate the effectiveness of dialect normalization, but do not directly measure whether semantically equivalent dialectal inputs lead to consistent model behavior. *Our work fills in this gap by introducing a unified robustness benchmark that evaluates LLMs across multiple tasks, including MCQA, NLI, and sentiment classification, covering five fine-grained regional dialect groups with human-annotated minimal pairs*

Contributions. Our main contributions are summarized as follows:

1. We introduce *Vialect-Bench*, the first high-quality, end-to-end human-annotated parallel benchmark for evaluating Vietnamese LLM robustness under meaning-preserving dialectal variation, covering five regional dialect groups and four language-understanding tasks (e.g., question answering, NER, NLI, sentiment classification).
2. We conduct a comprehensive empirical study of dialect-induced performance degradation across dialects, tasks, and modern LLMs.
3. [Minh Tran: *What result would change how people think about this problem?*]

2 Related Work

Vietnamese Dialect studies in Natural Language Processing. Vietnamese dialect processing has recently attracted increasing attention as regional variation challenges NLP systems trained mainly

on standard, Northern-based Vietnamese. Le and Luu (2023) introduce a Central–Northern dialect transfer corpus and show that commercial translation systems struggle with Central dialect expressions, while monolingual Vietnamese models such as BARTpho outperform multilingual models for dialect transfer. Their fine-tuned transfer modules also improve downstream machine translation and text-image retrieval without retraining full systems. Extending this line of work, Anh Ta et al. (2026) propose ViDia2Std, a nationwide dialect-to-standard corpus of over 13,000 sentence pairs from all 63 provinces, covering Central, Southern, and non-standard Northern varieties, and show that dialect normalization improves machine translation and sentiment analysis.

Prior work treats Vietnamese dialectal variation mainly as a preprocessing problem, showing that dialect-to-standard normalization can improve downstream systems. However, this does not reveal whether models are inherently robust to dialectal inputs: a model may perform well after normalization yet behave inconsistently on meaning-preserving dialectal variants. *Our work instead focuses on dialectal robustness evaluation, asking whether LLMs produce stable and correct predictions across semantically equivalent Vietnamese dialect forms. This enables controlled analysis of performance drops across tasks, dialect regions, and model families.*

Dialect Robustness of LLMs. Recent work has begun to benchmark LLM performance under dialectal variation across multiple languages and tasks.

(i) *English Dialect Benchmarks.* Ziems et al. (2022) introduce Multi-VALUE, a framework for cross-dialectal English NLP using transformations grounded in linguistic features across 50 English varieties. Lin et al. (2025b) present ReDial, which evaluates fairness and robustness on reasoning tasks using over 1,200 parallel Standard English–AAVE query pairs from benchmarks such as HumanEval and GSM8K, demonstrating broad LLM brittleness on AAVE reasoning inputs. Gupta et al. (2025) extend this to five underrepresented English dialects across 12 datasets, showing consistent performance drops on non-standard dialectal inputs.

(ii) *Multilingual and Non-English Benchmarks.* Faisal et al. (2024) present DialectBench, evaluating 281 language varieties across 40 language clusters and 10 NLP tasks. In Arabic, Mousi et al. (2025) introduce AraDiCE, covering Modern Standard Arabic and three dialects across multiple tasks.

Altakrori et al. (2026) adapt MMLU-style question answering into five Arabic dialects.

3 Dataset Construction

[Minh Tran: **Vialect-Bench** is a benchmark of more than ..., includes .. tasks — ??? —]

We first describe the data sources and sampling strategies in Section 3.1, and then present the dialect taxonomy in Section 3.2, and finally detail the annotation and quality assurance processes in Section 3.3.

[Minh Tran: Think about the criteria to construct the dataset: quality, accuracy,...]

3.1 Data Sourcing

3.2 Dialect Taxonomy

We selected six dialect groups representing major regional varieties of Vietnamese. Our goal is not to exhaustively represent all local dialects of Vietnamese, but rather to capture major regional variation that is expected to pose challenges for contemporary Vietnamese NLP systems. These groups were chosen based on linguistic analyses from prior work on Vietnamese dialectology (Hoang, 2009; Le and Luu, 2023; Anh Ta et al., 2026).

Group	Coverage
Northern Standard	Hanoi, Red River Delta
North-Central I	Thanh Hoa
North-Central II	Nghe An, Ha Tinh
North-Central III	Quang Binh, Quang Tri, Thua Thien Hue
South Central	Quang Nam–Binh Thuan
Southern	Ho Chi Minh City, Southeast, Mekong Delta

Table 1: Six dialect groups in Vialect-Bench.

Prior linguistic studies have documented systematic differences in lexicon, pronouns, and discourse markers across these dialect groups. Although Vietnamese is often broadly categorized into Northern, Central, and Southern dialect regions, considerable variation exists within the Central region. Consequently, we distinguish Thanh Hoa, Nghe-Tinh, and Binh-Tri-Thien as separate North-Central groups, while the South-Central group covers dialect varieties extending from Quang Nam to Binh Thuan, which share a number of characteristic lexical and phonological features documented in prior dialectological studies. Together with Northern Standard and Southern Vietnamese, these groups provide broad coverage of regional variation across

Vietnam. We focus on these regions for three reasons. First, Northern Standard Vietnamese serves as the dominant written standard and is the primary variety represented in existing Vietnamese NLP resources. Second, the Central dialect groups (Thanh Hoa, Nghe-Tinh, and Binh-Tri-Thien) remain relatively underrepresented despite exhibiting well-documented differences in lexicon, pronouns, and discourse markers compared to the standard variety. Third, the South-Central and Southern regions represent large speaker populations with distinctive linguistic characteristics, allowing us to evaluate the robustness of LLMs across a diverse range of geographically and sociolinguistically distinct Vietnamese varieties.

3.3 Annotation and Quality Assurance

We introduce a novel annotation pipeline designed to ensure meaning preservation, label consistency, and dialect authenticity cross six regional dialect groups and four language understanding tasks.

Annotator Recruitment. For each dialect group, we recruited four annotators, including one primary writer, two validators, and one senior adjudicator. All annotators were native speakers or long-term residents (≥ 10 years) of the target dialect region. Following Lin et al. (2025a), we did not impose hard constraints on dialect identity verification; annotators self-identified as proficient speakers of their respective dialects.

Guideline Construction. The dialect-specific annotation guidelines were derived from existing Vietnamese dialect resources (Hoang, 2009; Le and Luu, 2023; Anh Ta et al., 2026) and refined through consultation with native speakers from each dialect region. The guidelines specify allowed modifications (pronouns, dialect markers, particles) and forbidden modifications (sentence structure, task labels, named entities, numbers).

Qualification Phase. Before participating in full-scale annotation, all annotators completed a mandatory qualification phase consisting of three stages:

1. **Challenge Test:** Each annotator converted 10 standard Vietnamese samples into their target dialect, covering all four tasks (3 sentiment, 3 NLI, 2 MCQA, 2 NER).
2. **Cross-Review:** Other annotators from the same dialect group independently reviewed the converted samples using a 5-criterion rubric: mean-

ing preservation, label preservation, dialect authenticity, fluency, and guideline compliance

3. **Threshold:** Annotators who consistently failed to preserve meaning or task labels during the pilot phase were excluded from the main annotation process.

Annotation Workflow. Each standard Vietnamese instance underwent a three-stage review process:

1. **Primary Annotation:** A primary writer converted the standard sentence into the target dialect, following dialect-specific guidelines that specified allowed modifications (pronouns, dialect markers, particles) and forbidden modifications (sentence structure, task labels, named entities).
2. **Dual Validation:** Two validators independently reviewed the dialectal variant. Instances accepted by both validators were retained; instances rejected by both were returned for revision.
3. **Adjudication:** Disagreements between validators were resolved by a senior adjudicator, whose decision was final.

Quality Control Metrics. We implemented both human and automated quality checks:

- **Meaning Preservation Rate:** Percentage of samples where both validators agreed on meaning equivalence.
- **Annotation Acceptance Rate:** Ratio of accepted to total submitted annotations.
- **Inter-Annotator Agreement:** Percent agreement and Cohen’s κ between validators on acceptance decisions.

Annotation Statistics. This multi-stage pipeline was designed to promote annotation consistency, meaning preservation, and dialect authenticity across tasks and regions.

4 Experiments

[Minh Tran: Every experiment must serve one core idea, the central message for readers, the every section should support that: Dataset design: creates controlled dialectal variants

Eval: measures inconsistency and performance drop

Analysis: Explains which dialects/tasks/features cause failures]

[Minh Tran: A paper should feel ‘maximal’: strong paper should not leave obvious questions unanswered.

For our project, obvious reviewer questions would be: “Is this just normalization again?” “Are the dialect variants meaning-preserving?” “Are errors caused by dialect or by bad annotation?” “Does the benchmark cover enough dialect diversity?” “Are results consistent across model families?” “Does mitigation improve robustness or just translate everything to standard Vietnamese?”] We evaluate along four research questions.

- **RQ1:** Do LLMs preserve their predictions across meaning-equivalent Vietnamese dialectal variants?
- **RQ2:** Which dialect regions and linguistic phenomena cause the largest semantic decision flips?
- **RQ3:** When does dialectal variation become task-critical rather than merely surface-level noise?
- **RQ4:** Does dialect-to-standard normalization improve robustness, or does it sometimes erase task-relevant dialectal meaning?

Limitations

We discuss the limitations of our survey, including coverage gaps and rapidly evolving threat models.

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